

Board notes

Print this and carry it. The deck records what is *projected*; this records what gets *written*.

This session is not a board session. Every derivation is a split-surface step run with the picture advancing beside the algebra. The board carries the **ledger** — the lines the runs keep pointing back at, an hour after they were written — and the **axes**, drawn once and never wiped.

Conventions. **ASK** — put it to the room before writing, and the answer to expect. **POINT** — the moments you turn and point at the board rather than the screen. **IF ASKED** — the question that always comes.

The board plan

Screen down in the **centre**, a **left wing** and a **right wing**.

LEFT WING — the ledger. Five lines, written as they are derived, never wiped.

RIGHT WING — the (u, v) axes at minute 12, and the two vote tallies above them. Only the tallies get wiped, each when it is resolved.

Left wing, line 0 — minute 0, during the retrieval

$$\frac{du}{dt} = \frac{a}{1 + v^n} - u \qquad \frac{dv}{dt} = \frac{a}{1 + u^n} - v$$

Up before anyone writes anything, and up all period.

Left wing, line 1 — minute 12, the last step of Run 0

$$a = \frac{\alpha}{(\gamma + \mu) K}$$

Say the symbol collision once, here. On this line $\gamma + \mu$ is removal, from session 5, and the Hill coefficient is n , from session 4. Gardner's paper calls the two Hill coefficients β and γ ; the handout uses his letters, and there is no removal rate in it. Never put his γ and our $\gamma + \mu$ in one equation.

ASK: *three things in the lab change a — name them.* Promoter (up), degradation tag (down), operator affinity (up, through K). All three enter through one number.

Right wing — minute 12, the axes

Draw them once, large, and leave them: ***u* across, *v* up**. Nothing else goes on these axes; every figure today is this picture with something added, and the room should be able to look away from the screen and still see it.

DO NOT WIPE. You point at these at 21, 27, 36, 50 and 65.

Left wing, line 2 — minute 19, the end of Run 1

$$u = \frac{a}{1 + v^n} \qquad v = \frac{a}{1 + u^n}$$

Directly under line 0, labelled ***du/dt* = 0** and ***dv/dt* = 0**.

ASK before writing: *on the first curve, which variable is still?* Expect "both". The answer is ***u*** only — and that is the whole idea of a nullcline.

Right wing — minute 21, vote 1 tally

Four columns, A–D. Write the distribution. **Do not resolve it** — the resolution is the next surface and it is a picture, not an argument from you.

IF ASKED *which is right?* — "Draw the steep one and count."

Left wing, line 3 — minute 50, the last step of Run 3b

$$J = \begin{pmatrix} -1 & -g \\ -g & -1 \end{pmatrix} \qquad \lambda = -1 \pm g \qquad g \equiv \frac{n x^{n+1}}{a}$$

POINT at line 2 as you write *g*: it came from differentiating the first nullcline and using the second to clear the square.

ASK: *what is *g* in words?* A loop gain — how much of a nudge in ***v*** shows up in the rate of ***u***, at the fixed point. Session 7's positive autoregulation had the same condition as a slope.

IF ASKED *why is *J* symmetric?* — because both removal rates were scaled to 1 and the point is on the diagonal. Tag one repressor and neither holds. That is item 4 of the handout, and Thursday.

Left wing, line 4 — minute 58, the last step of Run 4. Box it.

$$a_c = n (n - 1)^{-(n+1)/n}$$

POINT at the date. Session 4, slide 13, three weeks ago, no proof. Say it.

Right wing — minute 65, vote 2 tally

Four columns again. Before wiping vote 1's tally, ask whether anyone moved. Expect B to be the majority: $n = 2$ is *not a threshold* is the misconception this session exists to kill, and the resolution surface shows the wall at $n = 1$.

What does not go on the board

- Any derivation. They are step runs with the picture beside them; that is the point of the rebuild.
 - The flow field, the eigenvector picture, the τ - Δ plane. You will make a mess of any of them in chalk and each is one click away.
 - The Gardner numbers. They are on the handout.
 - Anything during the handout. Circulate instead.
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After class

Photograph the left wing before it is wiped. The five lines are the session's summary and the first thing to check PS4 against.